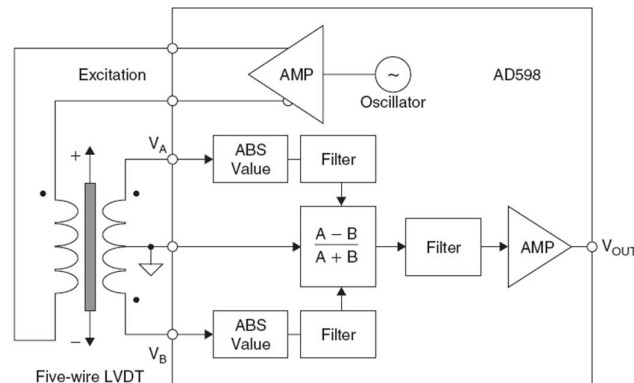


SCADA:

1. Better than HMI in basis of data processing and storage.
2. Monitor and control of individual machines of an entirety of automation factory.
3. Proper structuring of the system (i.e., designing) is a requisite for a SCADA.
4. Elements/variables used: i/p, o/p, reset value, accumulator, text display, driver selection and configurations.
5. Tag creation and configuration.

TRANSDUCER:

1. A **Linear Variable Differential Transformer (LVDT)** is a highly accurate, contact-less, electro-mechanical inductive transducer used to convert linear motion (position/displacement) into a proportional AC voltage signal. It consists of a primary coil and two secondary coils with a movable ferromagnetic core, offering high resolution and long life.



2. **Key Considerations for choosing a wiring configuration for Resistance Temperature Detectors:**
 - a. 2-Wire: Ideal for simple, low-power applications in hazardous or remote areas where minimal wiring is a priority. It is the most cost-effective option.
 - b. 3-Wire: A good balance between cost and performance for general industrial use. It is commonly used for sensors like RTDs where it compensates for lead wire resistance to improve accuracy over a 2-wire setup.
 - c. 4-Wire: Best for high-precision, critical applications, such as in laboratories or noisy industrial environments, where maximum accuracy and electrical isolation are essential and cost is less of a concern.
3. A **4-wire transmitter** is an industrial instrumentation device that uses two separate pairs of wires: one pair for its own power supply and the second pair for transmitting the signal (typically a 4-20 mA current loop) to a control system.
4. **Analog input for a Variable Frequency Drive (VFD)** enables modulation of motor speed based on external, continuous signals (typically 0-10V DC or 4-20mA) rather than fixed presets. This process involves a transmitter, signal conditioning (AMP/Tx), analog-to-digital conversion (ADC), and control via a PLC.
5. **Key Components in VFD Analog Input Loop**
 - a. Transmitter (Tx): Field devices (e.g., pressure sensors, level transducers, flowmeters) measure a process variable and convert it into a standard industrial signal, usually 4-20mA or 0-10V.
 - b. Amplifier/Signal Conditioner (AMP/Tx): Converts or boosts sensor signals to a level readable by the VFD or PLC, often isolating the signal to prevent noise interference.
 - c. Analog-to-Digital Converter (ADC): Internal to the VFD control card (or a PLC analog input card), it converts continuous analog voltage/current into digital values for internal processing.
 - d. PLC (Programmable Logic Controller): Processes sensor data and generates a 4-20mA or 0-10V output signal from an Analog Output (AO) card to the VFD to control speed.
 - e. VFD Input Terminals: Designated ports (e.g., AI1, AVI, ACI) on the drive that receive the signal.
6. **Typical Signal Flow & Configuration**
 - a. Sensor/Transmitter: Measures process value (e.g., Pressure: 0–10 Bar).
 - b. Signal Conversion: Transmitter sends 4–20 mA signal to VFD.

- c. VFD Configuration:
 - i. AI Terminal: Connected to AI2 & COM.
 - ii. Signal Type: Configured via parameters (e.g., AI2 set to 4–20mA).
 - iii. Scaling: Parameters map 4mA to minimum frequency (e.g., 0 Hz) and 20mA to maximum frequency (e.g., 50/60 Hz).
- d. Control Logic: The VFD's internal PID controller or speed reference logic receives this signal to adjust motor speed.

7. Common Analog Standards

- a. 4-20mA: Preferred in industrial environments due to high noise immunity and broken wire detection.
- b. 0-10V DC: Commonly used for direct connection to potentiometers or shorter cable runs.
- c. +/- 10 VDC: Used for bidirectional, forward/reverse motor control via a single analog channel.

8. Implementation Tips

- a. Isolation: Use isolated analog input modules to protect the PLC and VFD from surge voltages.
- b. Shielding: Use shielded, twisted-pair cables to prevent electrical noise interference from the motor.
- c. Grounding: Ensure the VFD, motor, and control signal cables are properly grounded to avoid ground loops.
- d. Parameter Settings: Ensure VFD parameters for AI signal type and scaling match the transmitter output.



- 9. **Analog output (AO)** is a critical interface for controlling the speed and operation of a Variable Frequency Drive (VFD) via a PLC, SCADA, or HMI system. This method provides smooth, proportional control rather than simple ON/OFF operation.

10. Key Components of Analog VFD Control

- a. VFD (Variable Frequency Drive): Receives the analog signal (typically 0-10V or 4-20mA) on its analog input terminals (e.g., AVI/ACI) to regulate frequency.
- b. PLC (Programmable Logic Controller): Uses an Analog Output module (e.g., Siemens S7-300, Delta DVP06XA-S2) to generate the signal, usually ranging from 0-10V or 4-20mA.
- c. HMI/SCADA: Provides the user interface to set, monitor, and adjust the frequency/speed, which is then mapped to the PLC register managing the AO.
- d. DAC (Digital-to-Analog Converter): The internal component of the PLC output module that converts the digital control word (e.g., 0-27648 in Siemens) into a physical voltage or current signal.

11. Servo Motor Requirements (FX5 can control 4 servos ATM):

A servo motor system must satisfy the following three basic requirements:

- 1) Direction Control,
- 2) Speed Control,
- 3) Position Control.

12. System Components and Wiring

The servo system must consist of a motion-capable PLC, servo amplifier, servo motor, power supply, and a dedicated communication interface. The PLC must support axis control functions, and the servo amplifier must be compatible with the selected control method. Electrical wiring for main power, control power, motor phases, encoder feedback, and communication must be completed and verified prior to software configuration.

13. Software Configuration

a. In GX Works3:

- i. Motion control must be enabled at project creation.

- ii. The servo amplifier is registered in the hardware or module configuration and assigned to a specific axis number.
 - iii. Axis parameters are configured, including operation mode, command unit system, electronic gear ratio, maximum speed, acceleration/deceleration, torque limits, and software stroke limits.
 - iv. Axis parameters in GX Works3 must be consistent with the mechanical design and servo amplifier settings.
- b. MR Configurator:**
 - i. Used for servo amplifier setup and verification.
 - ii. Motor model selection, control mode, communication settings, and encoder configuration are performed here.
 - iii. Servo alarms are diagnosed and reset, and gain tuning or auto tuning is executed to ensure stable motion performance.
 - iv. Parameter synchronization between MR Configurator and GX Works3 must be confirmed.
- 14. Operation Sequence and Safety**
 - a. Operation in GX Works3:**
 - i. Follows a defined control sequence.
 - ii. The servo enable signal is activated after confirming no active alarms and all safety conditions are satisfied.
 - iii. An origin return (homing) operation is executed to establish the mechanical reference position.
 - iv. After homing completion, motion commands such as absolute positioning, relative positioning, or velocity control are issued. Motion completion and error status are continuously monitored.
 - b. Safety Functions:**
 - i. Must be integrated into the control logic.
 - ii. Emergency stop, servo alarm detection, limit protection, torque monitoring, and interlock conditions must override motion commands.
 - iii. The servo must transition to a safe state immediately upon detection of a fault or safety violation.
 - iv. Proper coordination between PLC logic, GX Works3 motion configuration, and MR Configurator parameters is essential for reliable and safe servo system operation.
- 15. In GX Works3, the PLSV, DPLSV, PLSY, and DPLSY instructions are used for high-speed pulse output from the PLC's built-in I/O to control external devices like stepper or servo motors.**
 - a. **PLSV / PLSY:** These are 16-bit instructions, used when the pulse frequency and count can fit within a 16-bit data range.
 - b. **DPLSV / DPLSY:** These are 32-bit instructions, used when a larger number of pulses or a higher frequency is required, offering greater range and precision.
- 16. Instruction Format and Segments**

The general format for these instructions in ladder logic is [Instruction S1 S2 D]:

 - a. **S1 (Pulse Frequency):** Specifies the desired output frequency in Hz (Hertz).
 - i. This can be a constant (K value), a data register (D register), a timer (T), or a counter (C).
 - ii. The PLC will try to maintain this frequency for the duration of the pulse output.
 - b. **S2 (Number of Pulses):** Specifies the total count of pulses to be output.
 - i. This can be a constant (K value) or a data register (D register).
 - ii. Once this count is reached, the pulse output stops automatically until the instruction is triggered again.
 - iii. If S2 is set to 0, it indicates continuous pulse output (not limited by count).
 - c. **D (Pulse Output Terminal):** Specifies the physical PLC output point used for the pulses.
 - i. For smaller Mitsubishi PLCs, this is typically a high-speed transistor output such as Y0, Y1, or Y2.
- 17. Key Difference**
 - a. **PLSY / DPLSY** are for positioning control where a specific number of pulses are output to move to a defined position.
 - b. **PLSV / DPLSV** are for speed control, where the output is continuous based on the frequency specified in S1, effectively operating in a velocity control mode without a defined pulse count limit (often S2 is set to 0 in this case).
 - c. **DRVI / DDRVI** are for relative (incremental) positioning control.
 - d. **DRVA / DDRVA** are for absolute positioning control

18. DRVI / DDRVI Function Explained

The DRVI and DDRVI instructions are used to command a motor to travel a specified distance relative to its current position. This differs from absolute positioning commands, which move the motor to a fixed, global coordinate.

- a. **DRVI:** 16-bit instruction for standard pulse counts and frequencies.
- b. **DDRVI:** 32-bit instruction for larger pulse counts or higher frequencies, offering greater range.

19. Instruction Format and Segments

The general format for these instructions is:

20.	Instruction	S1	S2	D1	D2
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- a. **S1 (Pulse Frequency):** Specifies the speed of the movement in Hz. The movement will accelerate to and decelerate from this speed during the positioning move.
- b. **S2 (Number of Pulses):** Specifies the total number of pulses to output relative to the *starting position*. This value determines the total distance travelled.
- c. A **positive value** moves in the forward direction.
- d. A **negative value** moves in the reverse direction.
- e. **D1 (Pulse Output Terminal):** The physical output pin (e.g., Y0 or Y1) used for the pulses.
- f. **D2 (Direction Output Terminal):** The physical output pin (e.g., Y4 or Y5) that signals the servo drive which direction to move (ON for forward, OFF for reverse, or vice-versa depending on wiring/parameters).

This allows for easy forward and reverse positioning by simply changing the sign of the S2 value.

21. DRVA / DDRVA Function Explained

The DRVA and DDRVA instructions are used to command a motor to travel to a specific, fixed absolute position within the machine's coordinate system. This position is typically relative to the origin (home) position established by a homing operation.

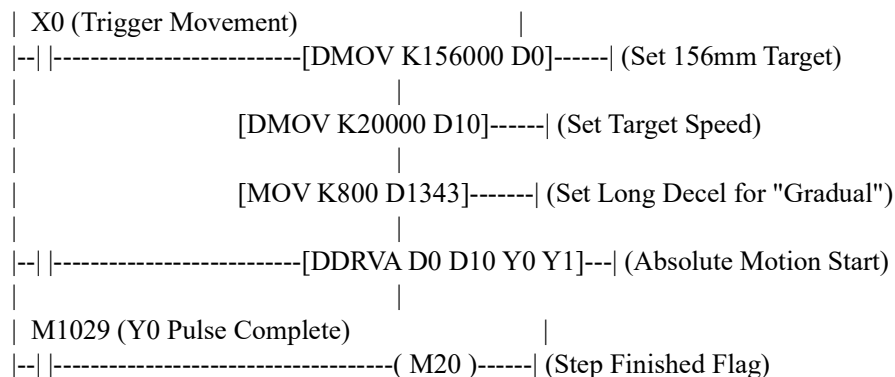
- a. **DRVA:** 16-bit instruction for standard frequency and position ranges.
- b. **DDRVA:** 32-bit instruction for extended frequency or larger position ranges.

22. Instruction Format and Segments

The general format for these instructions is identical to DRVI/DDRVI:

23.	Instruction	S1	S2	D1	D2
-----	--------------------	-----------	-----------	-----------	-----------

- a. **S1 (Pulse Frequency):** Specifies the speed of the movement in Hz.
 - b. **S2 (Absolute Position):** Specifies the exact target position value (in pulses from the home position).
 - c. **D1 (Pulse Output Terminal):** The physical output pin (e.g., Y0) used for the pulses.
 - d. **D2 (Direction Output Terminal):** The physical output pin (e.g., Y4) that signals the direction of travel.
- This instruction requires the system to have a properly executed origin return (homing) operation first so that the PLC knows where "Position 0" is before attempting to move to an absolute coordinate.



e. Configure Motion Parameters:

- i. **D1340:** Acceleration time (typically 100–500ms).
- ii. **D1343:** Deceleration time (determines how "gradual" the slow-down is).
- iii. **D1341:** Starting/Ending speed (bias frequency).